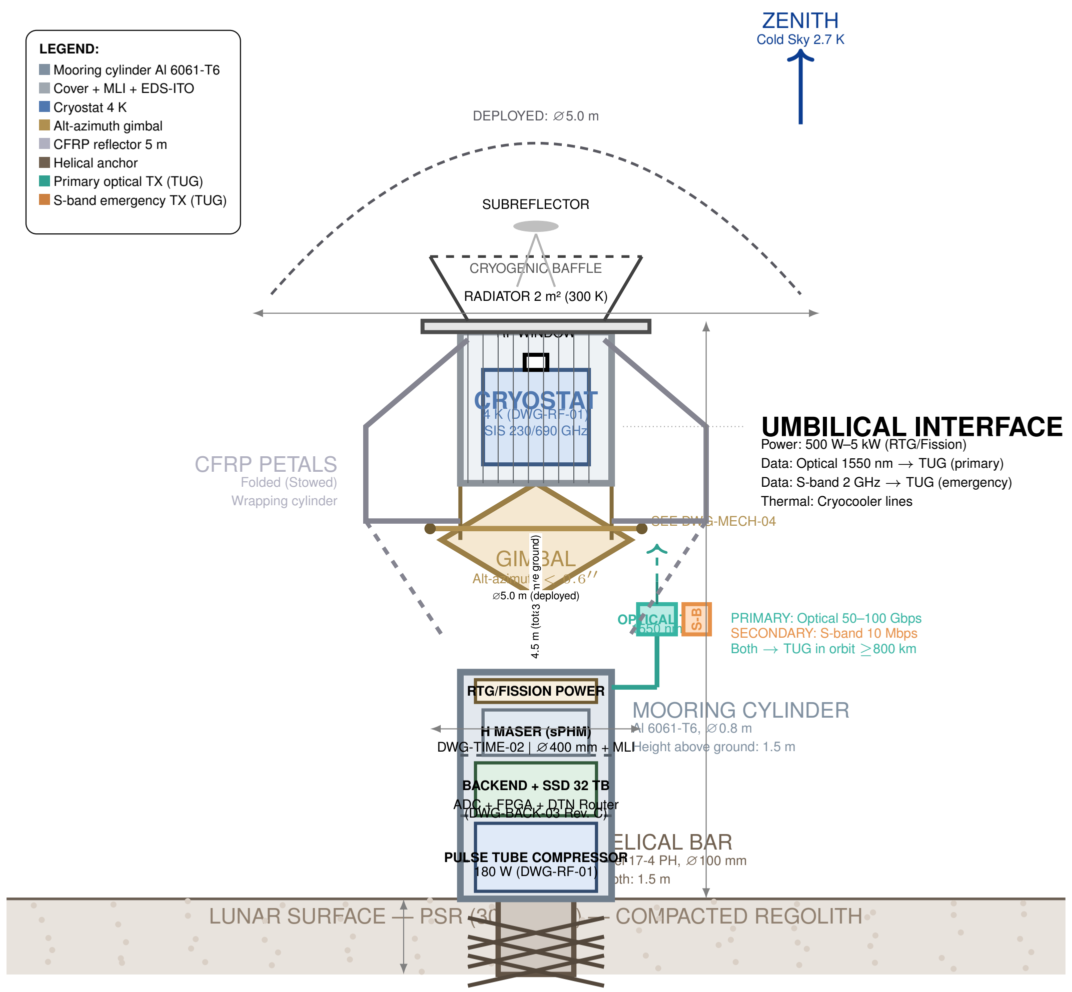
GENERAL ASSEMBLY — COMPLETE VLBI ANTENNA

VIEW 1: SIDE ELEVATION — LUNAR SURFACE TO ZENITH — DWG-ASSY-00 SHEET 1



MISSION:	L-RAIL (NASA/COPAG/PhysPAG)	REV: C
DRAWING:	DWG-ASSY-00 | SHEET 1 of 2	DATE: 2026-08-06
TITLE:	General Assembly — VLBI Antenna with Cover and Comms Module	SHEET 1/2
SCALE: 1:20	UNITS: mm / kg	FORMAT: A3 L

L-RAIL

GENERAL ASSEMBLY NOTES — DWG-ASSY-00

1. INSTALLATION SEQUENCE ON LUNAR SURFACE:
1.1. CLPS/ASLH lander descends in PSR (Lunar South Pole).
1.2. Helical drill deploys anchor bar and drills ∼1.5 m into regolith.
Deployment of ground radiation emission shielding and rakes for suspended regolith capture. 1.3. Mooring cylinder (∅0.8 m) descends along guide bar to surface.
1.4. Outer cover deploys (was folded in launch fairing).
1.5. Reflector petals open via kinematics DWG-DISH-03.
1.6. Lateral boom with communications module (optical TX + S-band) deploys.
1.7. Cryocooler starts; cryostat reaches 4 K in < 48 h.
1.8. Hydrogen maser stabilizes (additional 24 h).
1.9. Antenna operational: begins VCV48 phase calibration.

2. REGULATORY FRAMEWORK:
2.1. Bolted joints and hardpoints per **NASA-STD-5020**.
2.2. Deployment mechanisms per **NASA-STD-5017**.
2.3. Full assembly vibration testing: 14.1 g_{rms} per **MIL-STD-1540D**.
2.4. Pointing and radiation pattern verification per **IEEE Std 149**.
2.5. Electromagnetic compatibility per **MIL-STD-461G**.

3. ELECTROMAGNETIC COMPATIBILITY:
3.1. **PROHIBITED to use X-band (8 GHz)** as secondary link: falls within cryogenic IF band (4–12 GHz) and saturates HEMT/ADC.
3.2. Secondary link **S-band (2 GHz)**: far from IF (4–12 GHz) and from science band (230–690 GHz).
3.3. Optical TX 1550 nm: no RFI (optical band, does not interfere with RF).
3.4. Optical TX and S-band mounted on lateral boom, isolated from cryogenic cylinder.

4. DOCUMENT TRACEABILITY:
4.1. All subsystems have independent detailed drawings.
4.2. Mass budget of **519.1 kg** includes NASA Class B margin (20 %).
4.3. ASSY-800 updated to 26.0 kg: includes 32 TB SSD, zenith optical TX, optical gimbal, S-band and DTN router.
4.4. Communications architecture: (VLBI → TUG relay → Earth). See DWG-BACK-03 Rev. C.
4.5. Cylinder ∅0.8 m to accommodate sPHM maser (∅400 mm + MLI).

PSR ENVIRONMENTAL CONDITIONS

Parameter	Value
Ground temperature	30 K – 80 K
Ambient pressure	$< 10^{-10}$ Torr
Lunar gravity	1.62 m/s²
Cosmic radiation	0.5 Sv/yr
Regolith (charge)	Electrostatic ± 10 kV
Cold sky (zenith)	2.7 K

DIMENSIONAL SUMMARY

Parameter	Value
Total mass (Rev. C)	**519.1 kg**
Height above ground	4.5 m
Mooring cylinder diameter	0.8 m
Reflector diameter	5.0 m (deployed)
Mooring depth	1.5 m
Nominal power	∼460 W
Folded envelope	2.5 m × 1.8 m
Time to operational	< 72 h

COMMUNICATIONS MODULE

Link	Specification
Primary	Optical 1550 nm, 50–100 Gbps
Secondary	S-band 2 GHz, 10 Mbps
Destination	TUG-BIG / TUG-2 (≥800 km)
Protocol	DTN (CCSDS 734.2-B-1)
SSD buffer	32 TB (RAID 1, rad-hard)

GENERAL ASSEMBLY — COMPLETE VLBI ANTENNA

VIEW 2: MASS BUDGET AND BILL OF MATERIALS — DWG-ASSY-00 SHEET 2

CONSOLIDATED MASS BUDGET — NASA CLASS B

ID	Subsystem / Component	Mass (kg)	Margin (%)	Reference
ASSY-100	**Main Reflector** (Petals + Core + BUS + Hinges + LMM + EDS-ITO)	186.0	5	DWG-DISH-01/04
ASSY-200	**Subreflector + Struts** (CFRP Quadrupod + Cassegrain Mirror)	8.0	5	DWG-DISH-01
ASSY-300	**Cryostat + Front-End** (Vacuum chamber + Shields + SIS + Feedhorns + HEMT)	28.0	10	DWG-RF-01
ASSY-400	**Pulse Tube Cryocooler** (Cold head + Compressor + Ion pump)	23.5	10	DWG-RF-01
ASSY-500	**Radiator + Cryogenic Baffle** (2 m² panel + CFRP/MLI)	12.0	10	DWG-RF-01
ASSY-600	**Alt-Azimuth Gimbal** (Axes + 60/45 N·m motors + 24-bit encoders + Fork)	55.6	10	DWG-MECH-04
ASSY-700	**Atomic Clock (sPHM)** (Maser + Mu-metal shielding + Electronics + Synthesizer)	24.0	10	DWG-TIME-02
ASSY-800	**Backend + TUG Communications** (64 GHz ADC + FPGA + 32 TB SSD + DTN Router + Zenith Optical TX + Optical gimbal + Emergency S-band)	**26.0**	10	DWG-BACK-03 Rev. C
ASSY-900	**Outer Cover** (Al 6061 cylinder + MLI + EDS-ITO + RF Window)	18.0	10	DWG-ASSY-00
ASSY-1000	**Mooring Module** (Buried cylinder ∅0.8 m + 17-4 PH helical bar)	45.0	15	DWG-ASSY-00
ASSY-1100	**Wiring + Connectors** (Cryogenic RF + DC power + Data + Thermal)	8.0	20	All
			Subtotal (without system margin)	**432.6 kg**
			Global NASA Class B Margin (20 %)	**86.5 kg**
			TOTAL COMPLETE ANTENNA MASS (DEPLOYED)	**519.1 kg**

ASSY-800 at 26.0 kg (+8.0 kg). Communications to TUG:
32 TB SSD (3.2 kg) + zenith optical TX (3.5 kg) + optical gimbal (2.0 kg) + emergency S-band (0.8 kg) + DTN router (2.0 kg) + mounting boom (1.5 kg) + wiring (0.8 kg).
Total mass 519.1 kg (+2 %).

BILL OF MATERIALS
COMPONENTS

MISSION:	L-RAIL (NASA/COPAG/PhysPAG)	REV: C
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L-RAIL

ID	Description	Material	Mass (kg)	Flight Analog	Drawing Ref.
A-001	CFRP petals (6 pcs)	Cyanate Ester CFRP / M40J	65.0	ALMA Band 10	DWG-DISH-02
A-002	Rigid central core	CFRP + Invar 36	12.0	APEX	DWG-DISH-01
A-003	Rear truss (BUS)	Tubular CFRP + Ti-5Al-2.5Sn	5.0	JWST BSF	DWG-DISH-04
A-004	Piezoelectric actuators	PZT-5H / Invar 36	3.0	ALMA APC	DWG-DISH-04
A-005	Kinematic hinges (6 pcs)	Ti-6Al-4V + Si_3N_4	18.0	JWST	DWG-DISH-03
A-006	LMM mechanisms + bolts	TiNi / A286	6.0	Custom	DWG-DISH-03
A-007	EDS-ITO electronics	PCB + HV source	3.0	Custom	DWG-DISH-02
A-008	Cassegrain subreflector	CFRP + Aluminized/Gold	4.0	Herschel	DWG-DISH-01
A-009	Quadrupod arms	Unidirectional CFRP M40J	4.0	ALMA	DWG-DISH-01
A-010	Cryostat vacuum chamber	Al 6061-T6 + Indium seals	10.0	ALMA Band 9	DWG-RF-01
A-011	Thermal shields (40K/10K)	Cu OFHC + MLI	5.0	Herschel/SPIRE	DWG-RF-01
A-012	SIS mixers (2 pcs)	Nb/Al-AlOx/Nb + InP HEMT	2.5	ALMA Band 9	DWG-RF-01
A-013	Corrugated feedhorns (2)	Al 6061 + Gold coating	1.5	ALMA Band 9	DWG-RF-01
A-014	Pulse Tube cold head	Stainless steel 316L + Cu OFHC	8.0	PT-30 / SHI	DWG-RF-01
A-015	Remote compressor	Al 6061 + Electronics	14.0	PT-30 / SHI	DWG-RF-01
A-016	Ion pump + H_2 source	Steel/Ti + $LaNi_5$	1.9	Agilent / Custom	DWG-RF-01
A-017	Radiator 2 m²	Al honeycomb + OSR	8.0	Mars 2020	DWG-RF-01
A-018	Cryogenic baffle	CFRP + MLI	4.0	JWST	DWG-RF-01
A-019	Azimuth axis (Direct Drive)	Ti-6Al-4V + 60 N·m motor	19.8	ALMA nutator	DWG-MECH-04
A-020	Elevation axis (Direct Drive)	Ti-6Al-4V + 45 N·m motor	14.6	ALMA nutator	DWG-MECH-04
A-021	Fork arms + Platform	CFRP/Al 6061	10.5	Custom	DWG-MECH-04
A-022	24-bit encoders (2 pcs)	Glass/Invar 36	2.0	Renishaw	DWG-MECH-04
A-023	EM brake + Slip-ring	Steel/Ti + Fiber	1.8	Custom	DWG-MECH-04
A-024	Gimbal control electronics	Rad-Hard PCB	2.5	Custom	DWG-MECH-04
A-025	H maser cavity + Quartz	Cu OFHC + Fused quartz	5.3	GP-B sPHM	DWG-TIME-02
A-026	Mu-metal shielding (3 layers)	Mu-metal	3.2	GP-B	DWG-TIME-02
A-027	Maser control electronics	Rad-Hard PCB	5.0	Custom	DWG-TIME-02
A-028	Synthesizer + LO	GaAs/InP MMIC	2.0	ALMA LO	DWG-TIME-02
A-029	64 GHz ADC + FPGA + 32 TB SSD + DTN Router	GaAs/InP MMIC + Rad-Hard FPGA + NAND	18.0	EHT / ALMA PhE	DWG-BACK-03
A-030	Zenith optical TX 1550 nm + 2-axis gimbal (primary TUG)	InP/Erbium + Al 6061	5.5	DSOC / LCRD	DWG-BACK-03
A-030b	Emergency S-band TX 2 GHz + mounting boom (secondary TUG)	GaAs + Al 6061	2.3	Custom	DWG-BACK-03
A-031	Mooring cylinder ∅0.8 m	Anodized Al 6061-T6	25.0	Custom	DWG-ASSY-00
A-032	Helical anchor bar	Steel 17-4 PH	20.0	Custom	DWG-ASSY-00
A-033	Lunar protection cover	Al 6061 + MLI + EDS-ITO	18.0	Custom	DWG-ASSY-00
A-034	Cryogenic RF wiring	Cu/PEEK + Gore coax	3.0	ALMA	DWG-RF-01
A-035	DC power wiring	Cu/PEEK + Hermetic connectors	3.0	Custom	DWG-ASSY-00
A-036	Data/telemetry wiring	Rad-hard optical fiber	2.0	NASA LCRD	DWG-BACK-03
			Total Components (37 items)		**432.6 kg**

DIMENSIONAL SUMMARY — L-RAIL VLBI ANTENNA

Parameter	Description	Value
Total mass (deployed)	Includes 20 % NASA Class B margin	**519.1 kg**
Height above lunar surface	Cylinder base to baffle tip	4.5 m
Mooring cylinder diameter	∅0.8 m (Rev. B)	0.8 m
Deployed reflector diameter	Petals fully open	5.0 m
Mooring depth	Helical bar in regolith	3.0 m
Nominal operating power	Cryocooler + electronics + comms	$\sim$460 W
Folded envelope (fairing)	Height $\times$ diameter (stowed)	2.5 m $\times$ 1.8 m
Time to operational	From lunar landing to VLBI first light	< 72 h
Primary link	Optical 1550 nm $\rightarrow$ TUG ($\geq$800 km)	50–100 Gbps
Secondary link	S-band 2 GHz $\rightarrow$ TUG (emergency)	10 Mbps
Data buffer	Rad-hard SSD (RAID 1)	32 TB

Engineering drawing — DWG-ASSY-00 Rev. C. With TUG communications module (primary optical + secondary S-band), 32 TB SSD and DTN router. Architecture: VLBI $\rightarrow$ TUG relay $\rightarrow$ Earth. Cross-referenced with DWG-BACK-03 Rev. C, DWG-TUGBIG-00 Rev. D and DWG-TUG2-00 Rev. B. Ready for CDR.